

Harshit Agarwal

+91 8580506698 | harshitagarwal713@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Engineer focused on building and deploying real-world AI products. Experienced across full-stack and data workflows using Next.js, Python, PostgreSQL, PGVector, Pipecat, and n8n, with hands-on exposure to microservices, vector search, and cloud-based delivery. Passionate about practical AI engineering—integrating LLM-powered features, scalable architectures, and automated workflows to build products that solve real business problems at scale.

SKILLS

Languages: Python, JavaScript, SQL, CLI

Frontend: Next.js, React

Backend: REST APIs, SQLAlchemy

Databases: MySQL, PostgreSQL

AI & Systems: Vector Search (PGVector), Voice AI (Pipecat), Document AI & OCR, Workflow Automation (n8n), Microservices

Tools & Platforms: Git, GitHub, Postman, VS Code, Google Cloud Run

EXPERIENCE

Vaanii Global Technologies LLP

Jun 2026 – Present

Founding Software Engineer

Remote, India

- Own the end-to-end backend architecture for an AI-powered customer engagement platform, built independently from the ground up.
- Built a new **Voice AI backend** integrating LLM, Speech-to-Text, and Text-to-Speech providers into a unified conversation pipeline.
- Built a production **OCR-based document verification pipeline** for government documents, cross-verifying a submitted document against 10+ supporting documents at the field level.
- Achieved **99.9% text extraction accuracy** with table structure preserved, processing a complete 20-page verification set end-to-end in **33–40 seconds** across extraction, verification, and response.
- Designed CRM and campaign-management features (lead upload, status tracking, outbound voice campaigns) backed by asynchronous processing pipelines on **FastAPI**, **SQLAlchemy**, and **PostgreSQL**, deployed on **Google Cloud**.

ConversAI Labs

Jun 2025 – Jun 2026

Software Engineer (prev. Full Stack Intern)

Remote, India

- Led development of the **Pipecat-based Voice AI backend**, designed to scale to **900+ calls/hour**; handled **480+ calls/hour** in production under telephony provider concurrency limits.
- Built backend services with **FastAPI** and **SQLAlchemy**, and automated post-call workflows (WhatsApp messaging, form submission, CRM logging) end-to-end in n8n, removing manual lead processing.
- Reduced conversation latency and mid-call drop-offs by optimizing turn-taking logic in production.
- Built PGVector-based retrieval across 100+ transcripts to improve AI response accuracy.

PROJECTS

HuntLoop — AI-Powered Job Application Tracker | *Next.js, TypeScript, Supabase, PostgreSQL, Gmail API, LLMs*

- Built a full-stack job-application platform that tracks each application's status, resume version, emails, and follow-ups, with a **9-stage pipeline**, a Kanban board, and an event timeline.
- Built an **LLM routing layer** across multiple free-tier providers with automatic failover and rate-limit cooldowns, so a single provider's quota running out doesn't fail the request.
- Built a cover-letter generator that extracts structured data from job descriptions, picks the best-matching resume, and **flags claims the resume doesn't support** before the letter goes out.
- Integrated **Gmail over OAuth** to sync inbound email, match messages to applications, classify them with an LLM into user-approved status suggestions, and create follow-up drafts in the real inbox.
- Built job discovery across job boards and company careers pages, using **content hashing** to skip LLM calls on unchanged pages, plus a **Chrome extension** that saves a posting in one click.

EDUCATION

Krishna Engineering College

B.Tech in Computer Science

Aug 2022 – May 2025

Ghaziabad, Uttar Pradesh